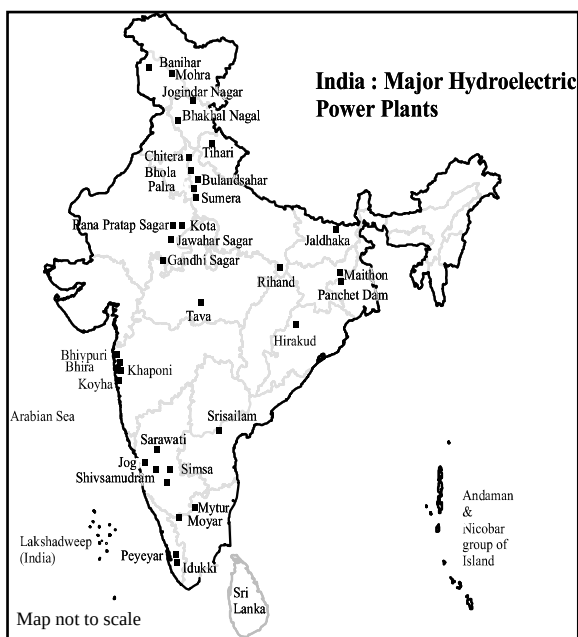




1. Where is the hydroelectric station located?

- (a) Koyna (b) Neyveli
(c) Kathgodam (d) Trombay

U.P.P.C.S. (Pre) 1992

Ans. (a)

The Koyna Hydroelectric Project is the largest hydroelectric power plant which is located on the Koyna, Maharashtra. Koyna dam, constructed on the River Koyna, is run by the Maharashtra State Electricity Board.

2. Rana Pratap Sagar Hydro Electricity Station is situated at :

- (a) Kota (b) Udaipur
(c) Rawatbhata (d) Bikaner

R.A.S./R.T.S. (Pre) 2007

Ans. (c)

Rana Pratap Sagar Hydel Power Station is located in the Rawatbhata town on the left side of the Rana Pratap Sagar Dam. It has been established with the cooperation of Canada.

3. Jawahar Sagar Hydroelectricity Project is situated on which river ?

- (a) Narmada (b) Chambal
(c) Tapi (d) Mahi

M.P.P.C.S. (Pre) 2020

Ans. (b)

Jawahar Sagar Hydroelectricity Project is situated on the Chambal River. It is related to the Chambal Valley Development Plan. This project is located in Rajasthan.

4. Rana Pratap Sagar Hydro-electricity Project on Chambal River is located in the State of:

- (a) Madhya Pradesh
(b) Rajasthan
(c) Uttar Pradesh
(d) Chhattisgarh
(e) None of the above / More than one of the above

66th B.P.C.S. (Pre) (Re-Exam) 2020

Ans. (b)

The Rana Pratap Sagar Dam is a gravity masonry dam, built on the Chambal River at Rawatbhat in Rajasthan.

5. What is the share of hydroelectric power in the total electricity produced in India?

- (a) 10 percent
(b) 12 percent
(c) 20 percent
(d) 22 percent
(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (a)

Hydro-electric power contributed to 10.6% of the total electricity produced during the question hour (2016-17). According to figures from the Ministry of Power the total installed capacity in India as of 31 December, 2024 was 462001.66 MW. The contribution of hydro-electric power in this is 46968.17 MW.

iv. Energy : Miscellaneous

*Sources of energy that appear in the form of coal, petroleum, natural gas and nuclear energy are commercial resources. They are also known as **non-renewable resources**. Whereas solar and wind energy, biogas, energy obtained from residue, geothermal and tidal energy, low hydroelectric and biomass energy are non-commercial and non-conventional sources. They are known as **renewable resource** of energy. Renewable resources or new resources are those resources which that replenished by natural / circumstantial Processes. *tge Ministry of New and Renewable Energy gave approval to the Consolidated Energy Policy on 26 December, 2008. *Solar energy is a huge resource of non-exhaustible form of energy. It is used for cooking, water purification, electricity

generation, etc. India receives approximately 5 kW per cm² of solar radiation energy for nearly 300 days in a year.

*Rampur village is the first village in India that has installed its own Solar Energy Plant and is situated in Uttar Pradesh. The first solar photovoltaic plant of 2MW capacity in India is installed in Bardhaman district of West Bengal near Asansol. Maharashtra is famous for the **Satara Wind Energy Plant**. The suitable states for the production of wind energy in India are Tamil Nadu, Andhra Pradesh, Gujarat, Karnataka, Kerala, Madhya Pradesh, Telangana and Rajasthan, etc.

*According to the figures of the Ministry of New and Renewable Energy the assessment indicates a gross power potential of 695.50 GW and 1163.85 GW in the country at 120 meter and 150 meters respectively, above ground level. The installed capacity of grid-interactive wind power in the country as of March 31, 2024 is 45.88 GW.

*According to the figures of the Ministry of New and Renewable Energy as of March 31, 2024, Gujarat (11722.72 MW) stands first the wind power installed capacity. followed by Tamil Nadu, Karnataka and Maharashtra.

State wise Wind Power Installed as on 31/03/2024 (MW)		
Sl. No.	State	Installed Capacity
1.	Andhra Pradesh	4096.65
2.	Gujarat	11722.72
3.	Karnataka	6019.61
4.	Kerala	63.5
5.	Madhya Pradesh	2844.29
6.	Maharashtra	5207.98
7.	Rajasthan	5195.82
8.	Tamil Nadu	10603.54
9.	Telangana	128.1
10.	Others	4.3
	Total	45886.51

*Electrical energy produced due to high and low tides in the sea is known as **tidal energy**. In India there are two major areas for Tidal Energy:

1. Rann of Kutch and Gulf of Khambhat (previously known as Gulf of Cambay) i.e. the Gujarat coastal area. The maximum tidal range is 8-11 m, whereas the average tidal range is 6.77 m-5.23 m. There is a tidal range of height 5 m approximately, whereas the average tidal range is 2.97 m.
2. Sundarban delta in West Bengal. India's first Geothermal Electric Plant was installed in **Khammam** in Andhra Pradesh (now Telangana) in 2012. The geothermal energy based **Manikaran Electrical Plant** is situated in Kullu, Himachal Pradesh. It is famous for its hot water sources. Electricity

produced from hot water sources is generated in the mountain valleys of Manikaran. Hot water sources are also found in Kasol, Kheer Ganga and Pulga in addition to Manikaran.

1. **Match List - I with List -II and choose the correct answer from the code given below :**

List - I (Site)	List -II (Energy)
A. Pavagada	i. Wind
B. Tattapani	ii. Solar
C. Muppandal	iii. Geo-thermal
D. Kakrapar	iv. Atomic

Code -

	A	B	C	D
(a)	iii	ii	i	iv
(b)	i	iii	ii	iv
(c)	i	ii	iii	iv
(d)	ii	iii	i	iv

R.A.S./R.T.S. (Pre) 2021

Ans. (d)

The correct match is as follows :

(Site)	(Energy)
Pavagada (Karnataka)	Solar
Tattapani (Chhattisgarh)	Geo - thermal
Muppandal (Tamil Nadu)	Wind
Kakrapar (Gujarat)	Atomic

2. **Which state has the highest potential for the development of solar energy?**

- (a) Madhya Pradesh (b) Jharkhand
(c) Uttar Pradesh (d) Rajasthan

Jharkhand P.C.S. (Pre) 2021

Ans. (d)

According to the Ministry of New and Renewable Energy, Government of India, Rajasthan has the highest potential for the development of solar energy (as of March 31, 2024).

3. **Match List-I with List-II and select the correct answer from the codes given below the lists:**

List- I (Power Plants)	List- II (Type of Energy)
A. Chamera	1. Atomic Power
B. Gwal Pahari	2. Thermal Power
C. Kudankulam	3. Hydro Power
D. Patratu	4. Solar Power

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	4	3	1

- (c) 4 3 2 1
(d) 3 4 1 2

U.P. B.E.O. (Pre) 2019

Ans. (d)

The Kudankulam nuclear power plant is located in Tamil Nadu. It was constructed in collaboration with Russia at Kudankulam in the Tirunelveli district. The Chamara power plant is a hydroelectric power plant located on the Ravi River in the Chamba district of Himachal Pradesh. The Patratu power plant is a thermal power plant in the Ramgarh district of Jharkhand State. There is a solar power plant on the Gwal Pahari (hill), located in Gurugram in Haryana. Hence, option (d) is the correct answer.

4. For which one of the following Satara is well known ?

- (a) Thermal Power Plant (b) Wind Energy Plant
(c) Hydro-electric Plant (d) Nuclear Power Plant

I.A.S. (Pre) 2005

Ans. (b)

Chalkewadi wind energy project in Satara (Maharashtra) is developed for generating electricity.

5. Which of the following states of India have the higher opportunities for the development of wind power?

- (a) Uttar Pradesh and Punjab
(b) Bihar and Jharkhand
(c) Tamil Nadu and Gujarat
(d) Rajasthan and Odisha

M.P.P.C.S. (Pre) 2020

Ans. (c)

According to the Ministry of New and Renewable Energy, Government of India, the total wind power capacity in the country is 695.50 GW at 120 meters above ground level and 1163.85 GW at 150 meters. According to this, the states (in GW) with more potential for the development of wind energy from the given options are as follows :

(State)	(120 meters above ground level)	(150 meters above ground level)
Gujarat	142.56	180.79
Rajasthan	127.75	284.25
Tamil Nadu	68.75	95.10

6. Which among the following ranks first in wind power generation?

- (a) Gujarat (b) Maharashtra
(c) Karnataka (d) Rajasthan

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (a)

In the context of wind power generation capacity (as of March 31, 2024), Gujarat is in the first position, followed by Tamil Nadu, which holds the second rank.

7. Which one of the following statements about energy production and consumption in India is not correct?

- (a) During the last decade, energy production in India has shown a declining trend.
(b) The per capita energy consumption in the world is the lowest in India
(c) The non-conventional sources of energy contribute less than one percent of the total commercial energy produced in India.
(d) The industry is the major energy consuming sector in India.

47th B.P.S.C. (Pre) 2005

Ans. (c)

The non-conventional sources of energy contribute approximately more than 1% of the total commercial energy produced in India. Thus, statement (c) is incorrect.

8. The biggest wind energy project in Asia with a capacity of 150 megawatts is located in :

- (a) Gujarat (b) Maharashtra
(c) Tamil Nadu (d) Karnataka

U.P.P.C.S. (Pre) 1999

Ans. (c)

At the time when the question was asked, Asia's biggest wind energy project was established in the village of Muppandal (150 MW), Tamil Nadu.

9. Which of the following coasts is the largest tidal energy producing area in India?

- (a) Mannar Coast (b) Kerala Coast
(c) North Circars Coast (d) Khambhat Coast

U.P.P.C.S. (Pre) 2022

Ans. (d)

Khambhat Coast is the largest tidal energy producing area in India.

10. Which one of the following is the main area for production "Tidal Energy"?

- (a) Gulf of Bengal
(b) Gulf of Mannar
(c) Gulf of Khambhat (Cambay)
(d) Gulf of Kutch

U.P.P.C.S. (Pre) 2012

Ans. (c)

India has a long coastline with estuaries and gulfs where tides are strong enough to move turbines for electrical power generation.

11. Most potential area for tidal energy in India is?

- (a) Madras
- (b) Cochin
- (c) Visakhapatnam
- (d) Bhavnagar

U.P.P.C.S. (Pre) 1991

Ans. (d)

India as a result of being surrounded by the sea on three sides has a high potential to harness tidal energy. The three most potential locations in this regard are Gulf of Cambay (west coast), the Gulf of Kutch (west coast) and Ganges Delta, Sundarbans, West Bengal (east coast). The most attractive locations are the Gulf of Cambay and the Gulf of Kutch.

12. Which of the following is related to the field of electricity generation?

- (a) MMTC
- (b) MTNL
- (c) NCL
- (d) NHPC

M.P.P.C.S. (Pre) 2012

Ans. (d)

NHPC is related to electricity generation. NHPC Limited (National Hydroelectric Power Corporation) is an Indian hydropower generation company that was incorporated in the year 1975 with the objective of planning, promoting and organizing an integrated and efficient development of hydroelectric power in all aspects. Later on, NHPC expanded its objectives to include other sources of energy like geothermal, tidal, wind, etc.

13. Which one of the following states of India ranks first in installed capacity and generation of electricity?

- (a) Andhra Pradesh
- (b) Gujarat
- (c) Maharashtra
- (d) Tamil Nadu

U.P.P.C.S. (Mains) 2006

Ans. (b)

When the question was asked option (b) was the correct answer. As of December 31, 2024, Gujarat ranks first in terms of gross electricity installed capacity.

14. In India, per capita consumption of energy in 1994 was –

- (a) 300 kg. of oil equivalent
- (b) 360 kg. of oil equivalent
- (c) 243 kg. of oil equivalent
- (d) 343 kg. of oil equivalent

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

In India, per capita consumption of energy in 1994 was an oil equivalent of 244 kg, of Oil which was much less than the world standard per capita consumption of energy of 1471 kg.

15. Rampura, which became the first village in India to have its solar power plant, is situated in –

- (a) Madhya Pradesh
- (b) Himachal Pradesh
- (c) Andhra Pradesh
- (d) Uttar Pradesh

U.P.U.D.A./L.D.A. (Pre) 2010

Ans. (d)

Rampura, which became the first village in India to have its solar power plant, is situated in Uttar Pradesh.

16. Consider the following reasons for the financial sickness of State Electricity Boards (SEBs) in India –

- I. Sale of power to agricultural and domestic consumers below the cost of production.**
- II. Transmission and distribution losses are very high.**
- III. Lack of commercial autonomy for SEBs.**
- IV. State Governments have implemented social subsidy policies through SEBs.**

Which of the above are correct?

- (a) I, II and III
- (b) I, II, III and IV
- (c) I, III and IV
- (d) II, III and IV

53rd to 55th B.P.C.S. (Pre) 2011

Ans. (b)

There are various loopholes in the functioning of State Electricity Boards in India and they have faced hardships due to administrative deficiencies, lack of financial resources, wide transmission and distribution losses, and a skewed tariff structure leading to unsustainable cross subsidies, etc. State Governments have implemented social subsidy policies through SEBs in the agrarian sector. According to the 11th five-year plan, India has higher transmission and distribution losses than other countries. The great deal of electricity in remote areas and less investment in the distribution system, etc., are prime roadblocks to these losses.

17. The power plant of Manikaran based on geothermal energy is in the State of

- (a) Arunachal Pradesh
- (b) Himachal Pradesh

- (c) Jammu and Kashmir
(d) Uttarakhand

Jharkhand P.C.S. (Pre) 2013

Ans. (b)

The power plant of Manikaran based on geothermal energy is located in the Parvati Valley on the river Parvati, northeast of Bhuntar in the Kullu district of Himachal Pradesh.

18. Which of the following is not correctly matched?

Hot springs	Location
(a) Manikaran	- Himachal Pradesh
(b) Jwalamukhi	- Jammu and Kashmir
(c) Anihoni	- Madhya Pradesh
(d) Tapta Pani	- Odisha

U.P.P.C.S. (Pre) (Re-Exam) 2015

Ans. (b)

Jwalamukhi is a town and municipal council in Kangra district of Himachal Pradesh. Thus, option (b) is not correctly matched.

19. Which of the following sources of energy is not a commercial source?

- (a) Petroleum (b) Atomic Energy
(c) Natural Gas (d) Biogas

U.P.R.O./A.R.O. (Mains) 2013

Ans. (d)

Coal, petroleum, natural gas and atomic energy are commercial sources of energy, also known as non-renewable and traditional resources, whereas wind, solar, biomass, geothermal, tidal and hydropower energy sources are non-commercial sources of energy, also known as renewable non-traditional resources.

20. Which is not a renewable resource ?

- (a) Energy of water
(b) Energy of Sun
(c) Energy of Earth
(d) None of the above

M.P.P.C.S. (Pre) 2015

Ans. (d)

Renewable sources are those that are replenished by the environment over a relatively short period of time. These types of resources renew so quickly that they are regenerated by the time we use them completely. Solar energy, the energy of water, the energy of the Earth and wind energy are all renewable sources of energy. Thus, option (d) is the correct answer.

21. Which among the following statements are true about 'Urja Ganga' project?

1. It is a gas pipeline project
2. It was launched in October 2016
3. It runs from Iran to India

Choose the correct answer from the code given below:

Codes :

- (a) Only 2 and 3 are correct
(b) Only 1 and 2 are correct
(c) Only 1 and 3 are correct
(d) All 1, 2 and 3 are correct

U.P.P.C.S. (Pre) 2017

Ans. (b)

On 24 October 2016, Prime Minister Narendra Modi laid the foundation stone of the Urja Ganga gas pipeline project in Varanasi.

22. The first Union Territory of India to run 100 percent on solar energy is :

- (a) Chandigarh
(b) Diu
(c) Andaman - Nicobar
(d) Puducherry

U.P.R.O./A.R.O. (Pre) 2017

Ans. (b)

India's first Union Territory was formed. The geographical area of Diu is approximately 40 sq. km. With effect from January 26, 2020, Daman and Diu and Dadra and Nagar Haveli were merged into one Union Territory.

23. Which one of the following states is a leading producer of solar energy in India?

- (a) Telangana
(b) Karnataka
(c) Andhra Pradesh
(d) Rajasthan
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

Rajasthan (as on March 31, 2024) tops the list of states as a leading producer of solar power in the country.